

CFE230

Basic computer knowledge and computer evolution



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Keywords:

Computer evolution; basic computer knowledge; Mathematical calculation in computer

Objectives

- To tell the computer development from mechanical calculator to the computer system.
- to tell difference between microprocess and microcontroller.
- To express electronic devices and electronic measurement tools
- Be able to describe the concept of data transferring and detection errors in the computer system.
- Be able to tell mechanical protection and safety in the computer system and IoT

Topic

- From Mechanical machine to Computer
 - Mechanical
 - Vacuum tube
 - Silicon Transistor
- Microprocessor, microcontroller, DSP, GPU
- Quiz (5 points)

1.1.1 Mechanical computer

- Abacus
 - Used in China, Europa and Russia

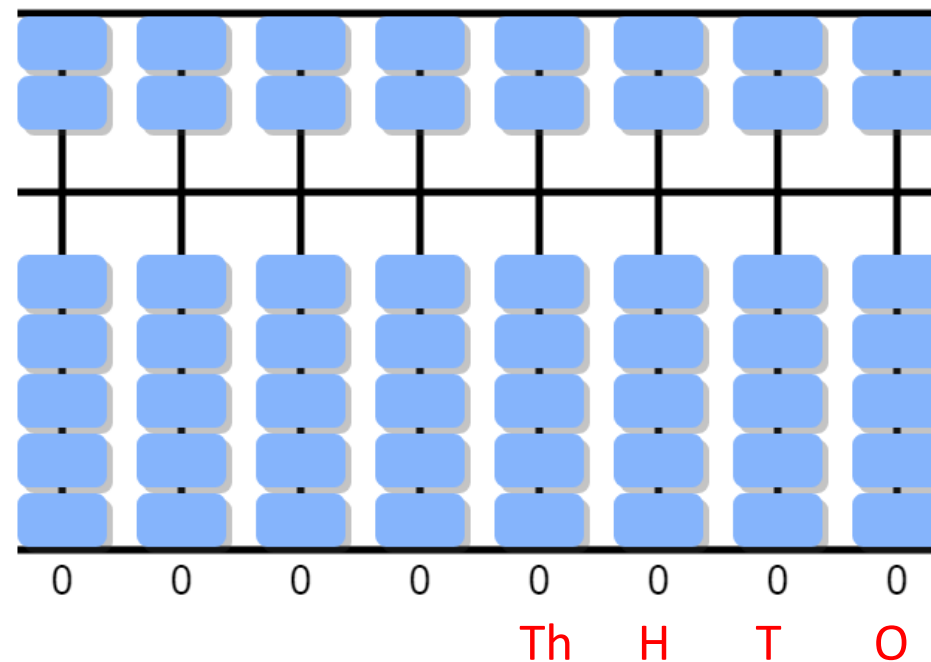


Chinese abacus

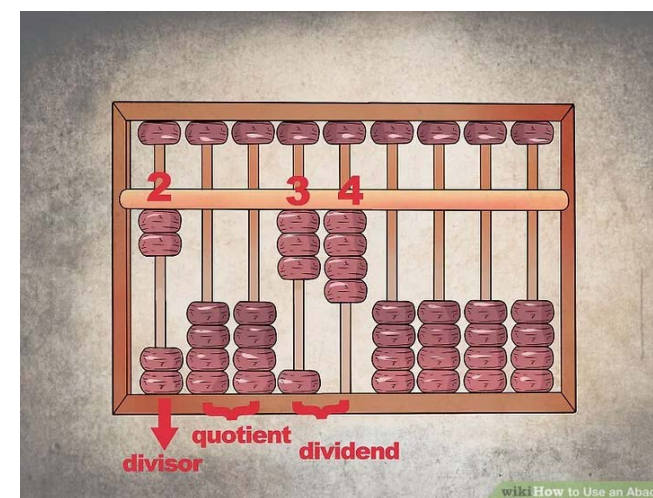
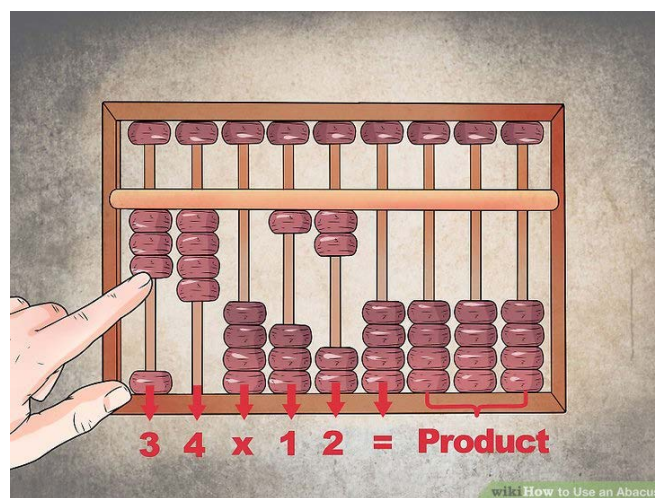
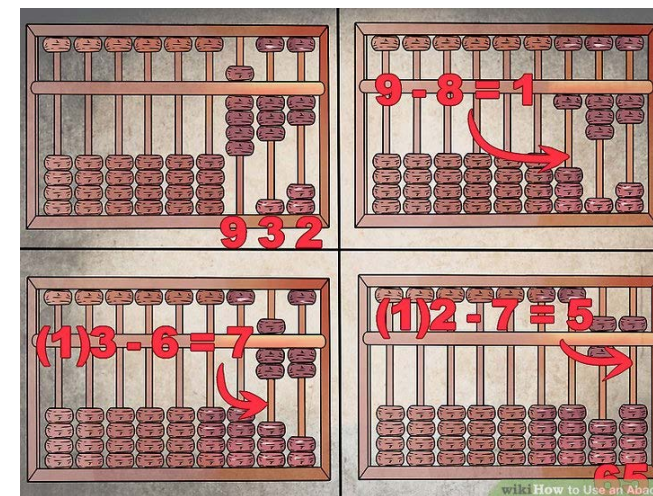
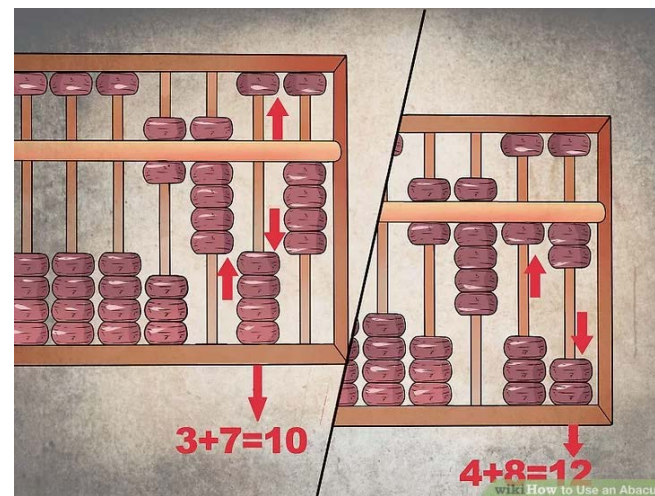
Activity 1.1 Play abacus

- Click on <https://www.mathematik.uni-marburg.de/~thormae/lectures/ti1/code/abacus/sanpan.html>

- Calculate
 - $83 + 12 = ?$
 - $112 + 12 = ?$



More functions in Abacus



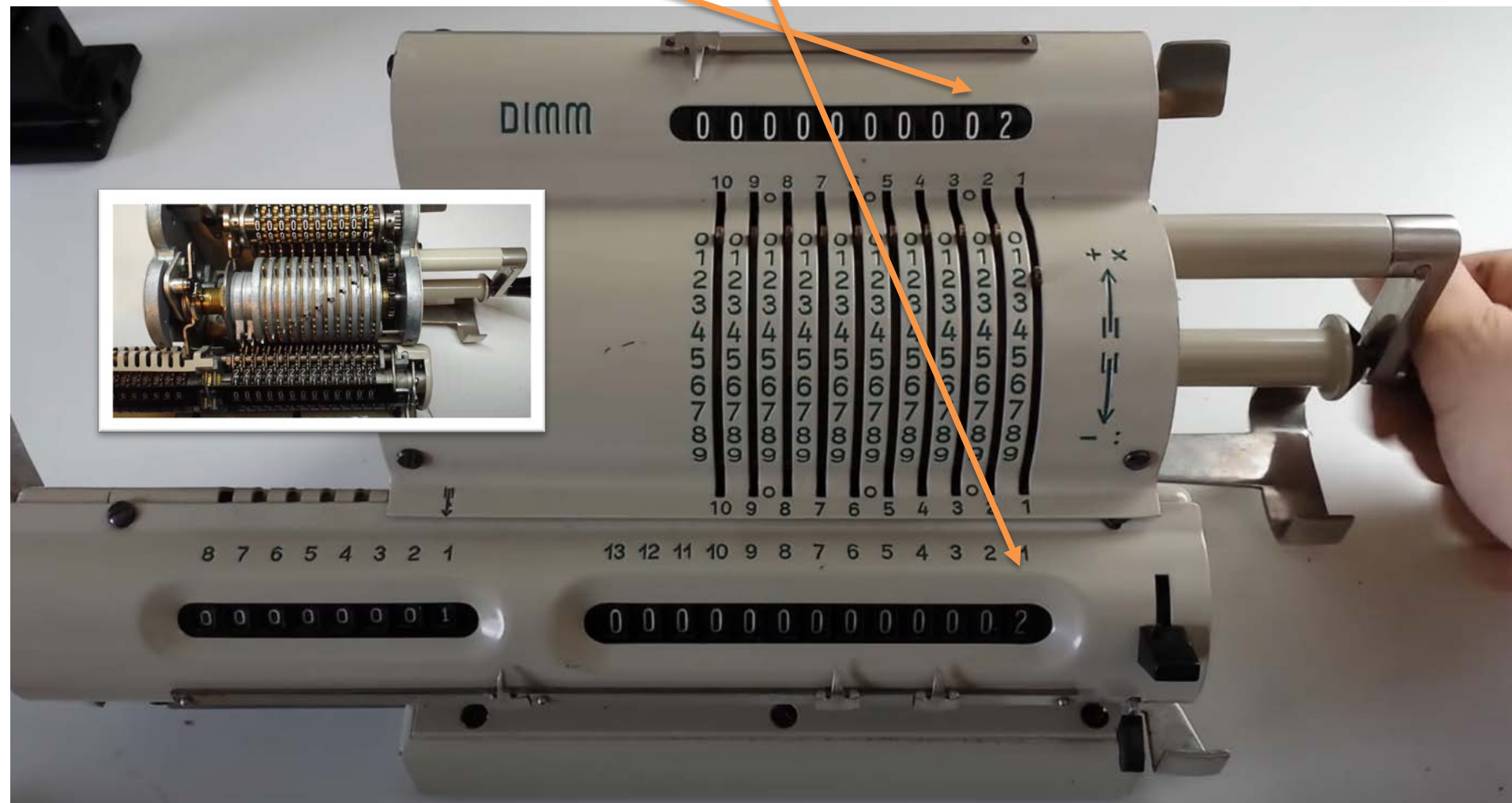
<https://www.wikihow.com/Use-an-Abacus>

1.1.1 Mechanical computer



- Mechanical calculator
 - Blaise Pascal (1642) developed a machine to help his father working in the shop.
 - Working only function of the addition and subtraction.

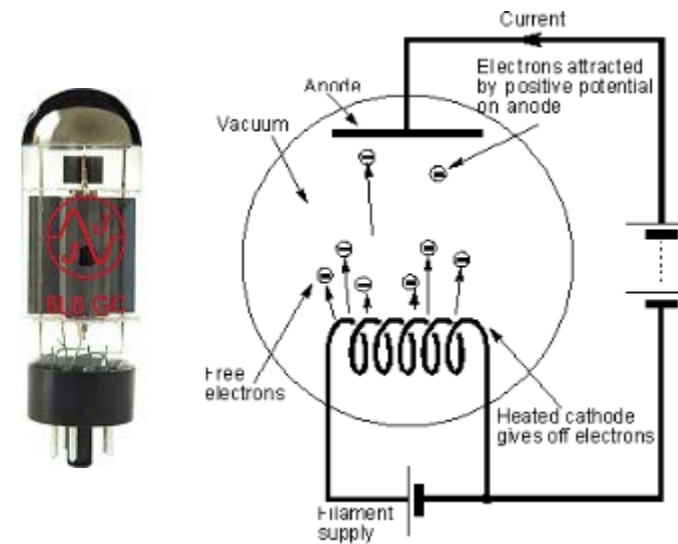
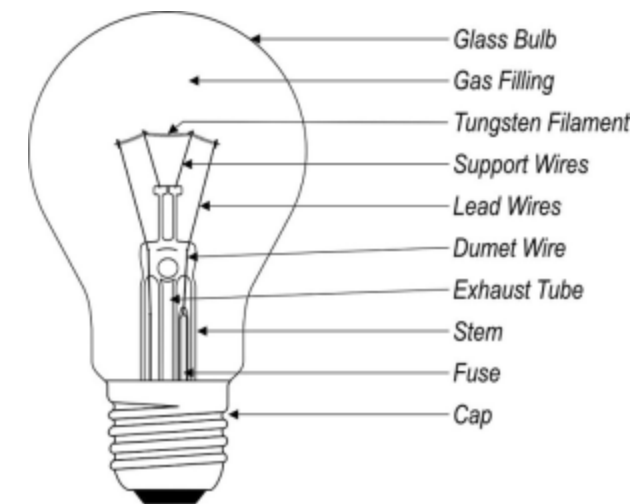
$$2+2 = 4$$

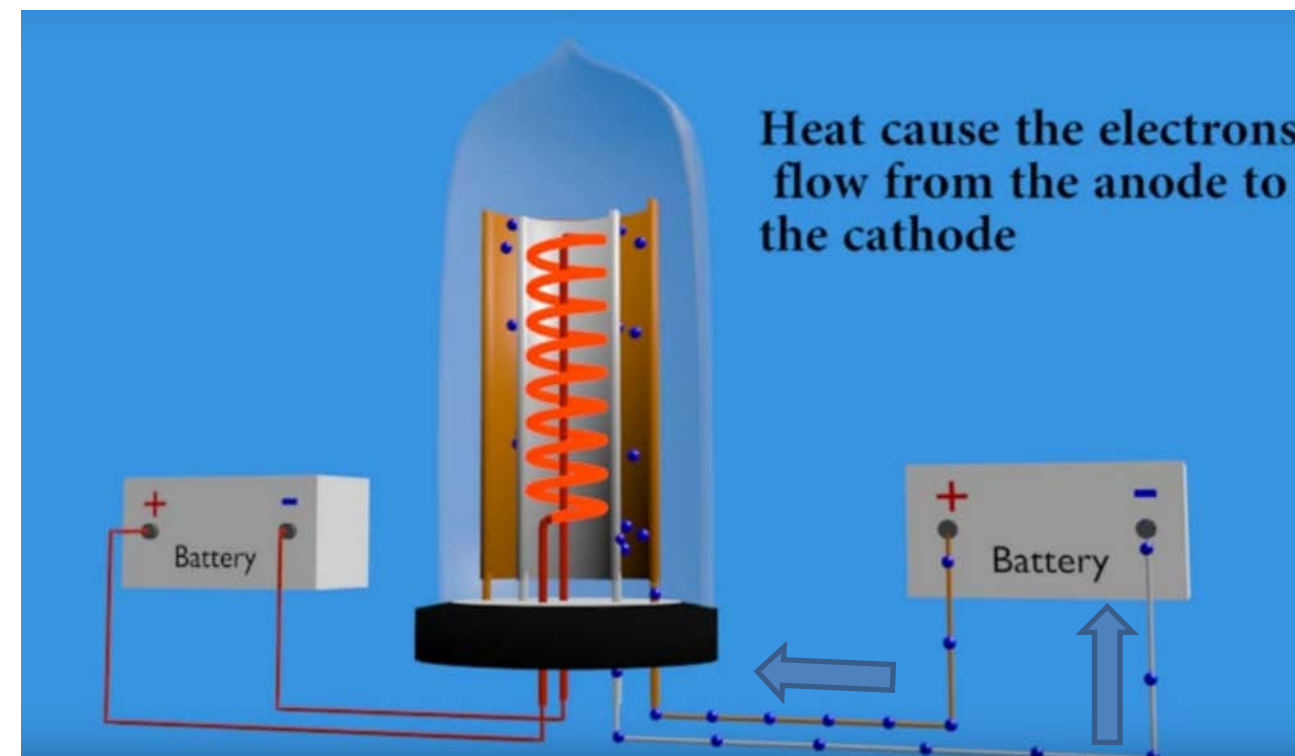
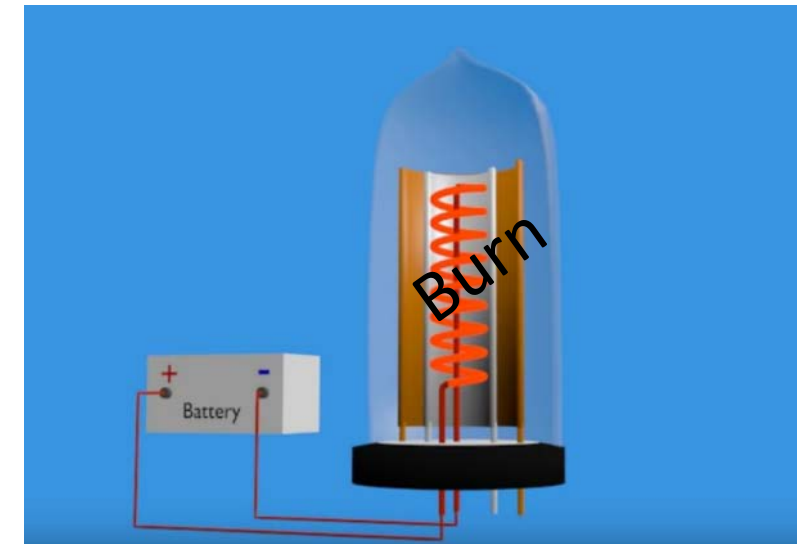
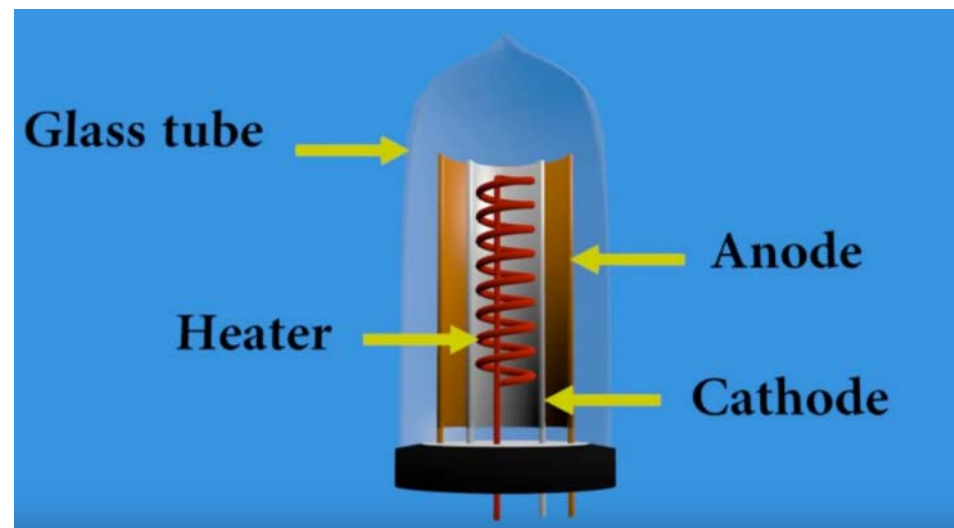


<https://youtu.be/aDN4s8ElxqE?t=144>

1.1.2 Vacuum Tube

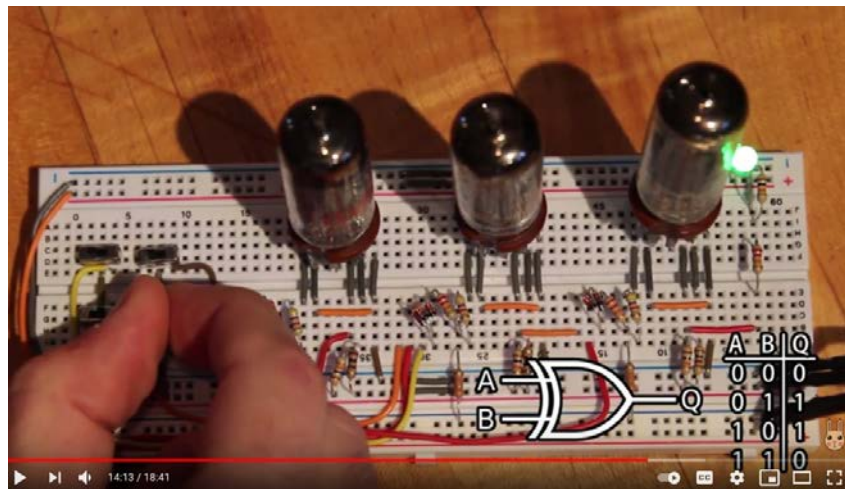
- Vacuum Tube was developed from *Incandescent lamp*.
- When a *tungsten*⁷⁴ lead gets heat, it spreads electron by moving to an anode plate.
- A grid plate uses controls number of electron moving from the tungsten lead to an anode plate.





<https://www.youtube.com/watch?v=K6BgZ8s1VuW>

Vacuum tube and future

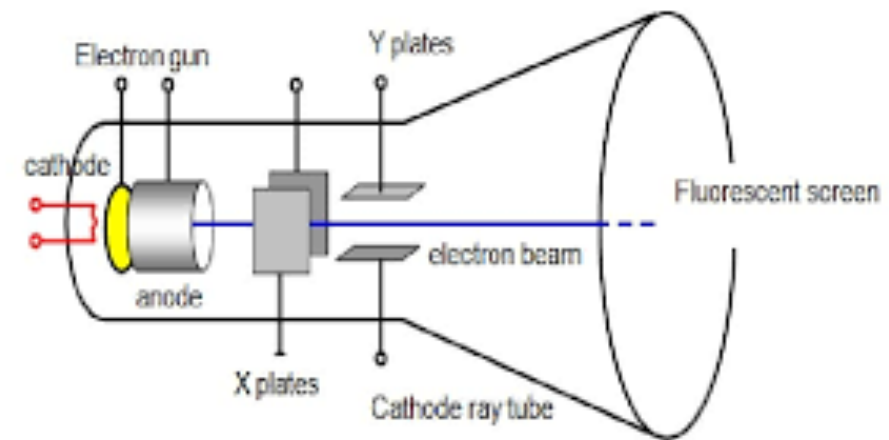
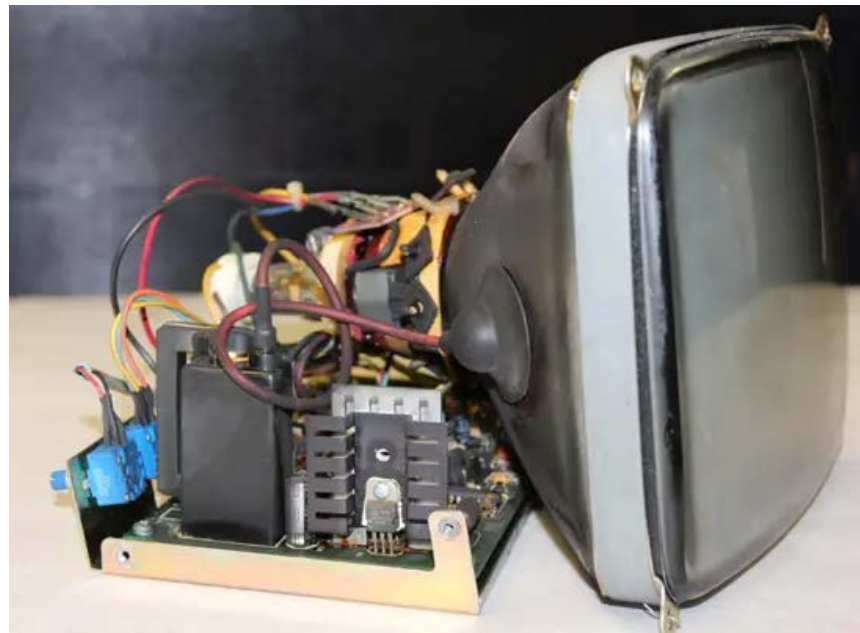


https://youtu.be/N-Sc6k_rITM?t=853

4.) Future: The Nano Vacuum Tube

Vacuum tubes may make a comeback and replace standard microchips. Engineers have been able to build a structure in phosphorus doped silicon and use nanotubes to build a switch. These devices can operate 10 times faster than silicon transistors.

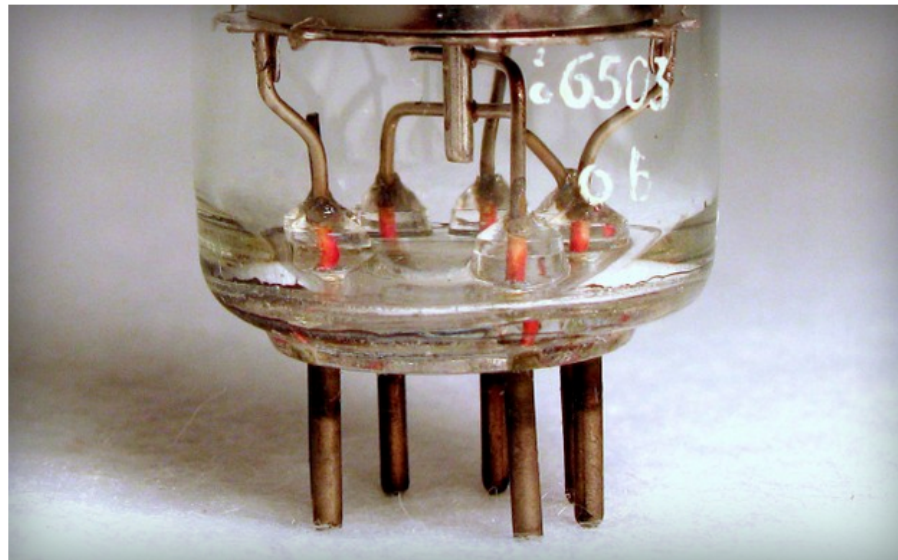
Cathod Ray Tube (CRT)



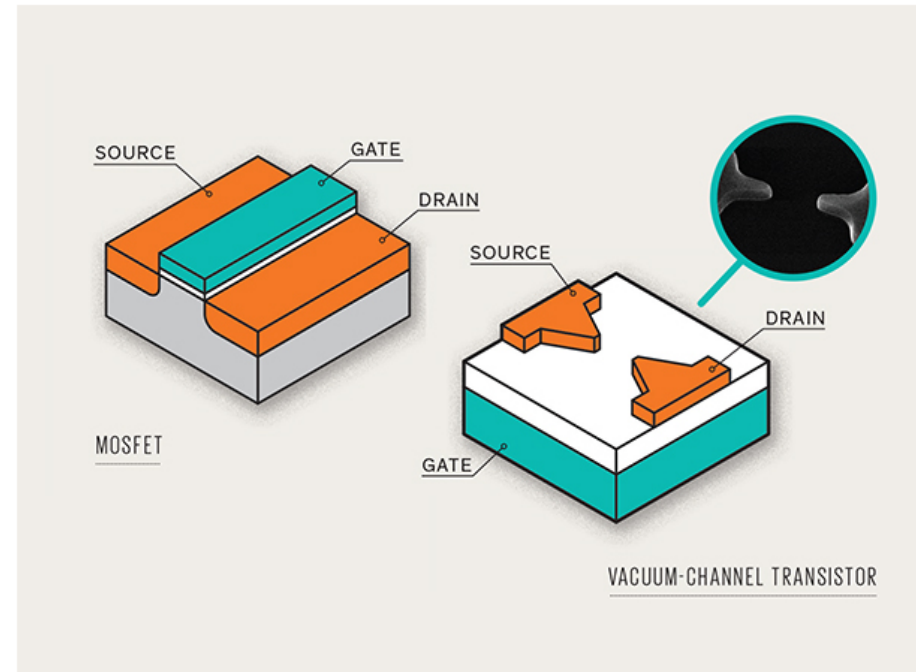
The vacuum tube strikes back: NASA's tiny 460GHz vacuum transistor that could one day replace silicon FETs

By Sebastian Anthony on June 24, 2014 at 8:12 am | [Comments](#)

[f](#) [t](#) [G+](#) [v](#) [Y](#) [F](#) 92 SHARES



Way back in the salad days of digital computing (the 1940s and '50s), computers were made of vacuum tubes — big, hot, clunky devices that, when you got right down to it, were essentially glorified light bulbs. This is why early computers like the ENIAC weighed more than 27 tons and consumed more power than a small town. Later, obviously, vacuum tubes would be replaced by probably the greatest invention of all time — the solid-state transistor — which would allow for the creation of smaller, faster, cheaper, and more reliable computers. Fast forward to 2014, though, and the humble CMOS field-effect transistor (FET) is starting to show its age. We've pretty much hit the limit on shrinking silicon transistors any further, and they can't operate at speeds much faster than a few gigahertz. Which is why NASA's Ames Research Center is going back to the future with its new *vacuum transistor* — a nanometer-scale vacuum tube that, in early testing, has reached speeds of up to **460GHz**.

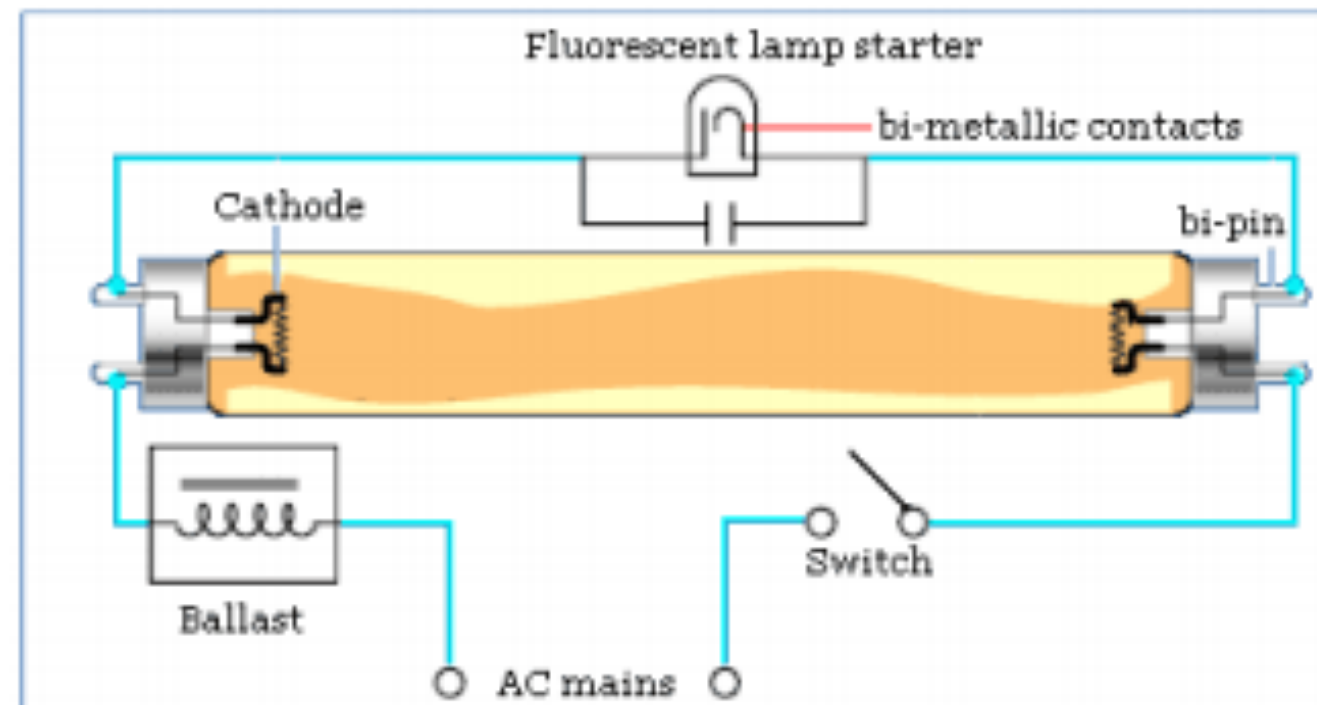


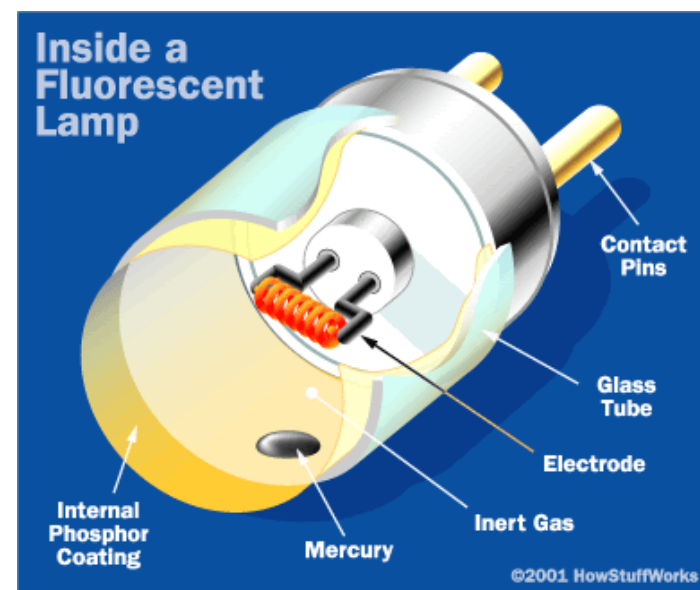
Standard MOSFET vs. vacuum-channel transistor [Image credit: IEEE Spectrum]

NASA's Ames Research Center has been working on-and-off for many years on the **vacuum-channel transistor**, which is essentially a vacuum tube that can be fabricated using conventional CMOS techniques. Instead of a gate sitting between the source and drain, there is... nothing. A vacuum. By a method known as *field emission*, electrons are drawn across the vacuum from the source to the drain when a current is applied to the gate (see diagram). By using field emission rather than the thermionic (hot) electron emission, these vacuum-channel transistors don't require a heat source. Importantly, they also don't require a vacuum — instead they use helium (it's sparse enough that the electrons have almost no chance of bumping into helium atoms while they traverse the few-nanometer gap between source and drain). The electrons also traverse the air gap a *lot* faster than if they had to pass through the gate electrode.

Activity1.2

- The fluorescence lamp has two small tungsten coils. Do you know why does the fluorescence lamp generates light? (Discussion)





1.1.3 Silicon Transistor

- Where are Silicon, Germanium, Boron, and Antimony on the periodic table?

1
1A
1
H
Hydrogen
1.008
1

2
2A
4
Be
Beryllium
9.012
4

3
3A
6
Li
Lithium
6.941
3

4
4A
8
B
Boron
10.81
5

5
5A
10
C
Carbon
12.011
6

6
6A
12
N
Nitrogen
14.007
7

7
7A
14
O
Oxygen
15.999
8

8
8A
16
F
Fluorine
18.998
9

9
9A
18
Ne
Neon
20.180
10

11
11A
20
Na
Sodium
22.990
11

12
12A
22
Mg
Magnesium
24.305
12

13
13A
24
Al
Aluminum
26.982
13

14
14A
26
Si
Silicon
28.086
14

15
15A
28
P
Phosphorus
30.974
15

16
16A
30
S
Sulfur
32.06
16

17
17A
32
Cl
Chlorine
35.45
17

18
18A
34
Ar
Argon
39.948
18

19
19A
36
K
Potassium
39.098
19

20
20A
38
Ca
Calcium
40.078
20

21
21A
40
Sc
Scandium
44.956
21

22
22A
42
Ti
Titanium
47.88
22

23
23A
44
V
Vanadium
50.942
23

24
24A
46
Cr
Chromium
51.996
24

25
25A
48
Mn
Manganese
54.938
25

26
26A
50
Fe
Iron
55.845
26

27
27A
52
Co
Cobalt
58.933
27

28
28A
54
Ni
Nickel
58.69
28

29
29A
56
Cu
Copper
63.546
29

30
30A
58
Zn
Zinc
65.38
30

31
31A
60
Ga
Gallium
69.723
31

32
32A
62
Ge
Germanium
72.63
32

33
33A
64
As
Arsenic
74.922
33

34
34A
66
Se
Selenium
78.96
34

35
35A
68
Br
Bromine
79.904
35

36
36A
70
Kr
Krypton
83.798
36

37
37A
72
Rb
Rubidium
85.468
37

38
38A
74
Sr
Strontium
87.62
38

39
39A
76
Y
Yttrium
88.906
39

40
40A
78
Zr
Zirconium
91.224
40

41
41A
80
Nb
Niobium
92.906
41

42
42A
82
Mo
Molybdenum
95.94
42

43
43A
84
Tc
Technetium
98
43

44
44A
86
Ru
Ruthenium
101.07
44

45
45A
88
Rh
Rhodium
102.91
45

46
46A
90
Pd
Palladium
106.42
46

47
47A
92
Ag
Silver
107.87
47

48
48A
94
Cd
Cadmium
112.41
48

49
49A
96
In
Indium
114.82
49

50
50A
98
Sn
Tin
118.71
50

51
51A
100
Sb
Antimony
121.76
51

52
52A
102
Te
Tellurium
127.6
52

53
53A
104
I
Iodine
126.91
53

54
54A
106
Xe
Xenon
131.29
54

55
55A
108
Cs
Cesium
132.91
55

56
56A
110
Ba
Barium
137.33
56

57-71
57-71
Lanthanides

58
58A
112
Hf
Hafnium
178.49
58

59
59A
114
Ta
Tantalum
180.948
59

60
60A
116
W
Tungsten
183.84
60

61
61A
118
Re
Rhenium
186.21
61

62
62A
120
Os
Osmium
190.23
62

63
63A
122
Ir
Iridium
192.22
63

64
64A
124
Pt
Platinum
195.08
64

65
65A
126
Au
Gold
196.97
65

66
66A
128
Hg
Mercury
200.59
66

67
67A
130
Tl
Thallium
204.38
67

68
68A
132
Pb
Lead
207.2
68

69
69A
134
Bi
Bismuth
208.98
69

70
70A
136
Po
Polonium
209
70

71
71A
138
At
Astatine
210
71

72
72A
140
Rn
Radon
222
72

73
73A
142
Fr
Francium
223
73

74
74A
144
Ra
Radium
226
74

75-103
75-103
Actinides

76
76A
146
Rf
Rutherfordium
261
76

77
77A
148
Db
Dubnium
262
77

78
78A
150
Sg
Seaborgium
266
78

79
79A
152
Bh
Bohrium
264
79

80
80A
154
Hs
Hassium
277
80

81
81A
156
Mt
Meitnerium
268
81

82
82A
158
Ds
Darmstadtium
271
82

83
83A
160
Rg
Roentgenium
272
83

84
84A
162
Cn
Copernicium
285
84

85
85A
164
Nh
Nihonium
284
85

86
86A
166
Fl
Flerovium
289
86

87
87A
168
Mc
Moscovium
288
87

88
88A
170
Lv
Livermorium
293
88

89
89A
172
Ts
Tennessine
289
89

90
90A
174
Og
Oganesson
294
90

91
91A
176
La
Lanthanum
138.91
91

92
92A
178
Ce
Cerium
140.12
92

93
93A
180
Pr
Praseodymium
140.91
93

94
94A
182
Nd
Neodymium
144.24
94

95
95A
184
Pm
Promethium
145
95

96
96A
186
Sm
Samarium
150.36
96

97
97A
188
Eu
Europium
151.96
97

98
98A
190
Gd
Gadolinium
157.25
98

99
99A
192
Tb
Terbium
158.93
99

100
100A
194
Dy
Dysprosium
162.50
100

101
101A
196
Ho
Holmium
164.93
101

102
102A
198
Er
Erbium
167.26
102

103
103A
200
Tm
Thulium
168.93
103

104
104A
202
Yb
Ytterbium
173.05
104

105
105A
204
Lu
Lutetium
174.97
105

106
106A
206
Ac
Actinium
227
106

107
107A
208
Th
Thorium
232.04
107

108
108A
210
Pa
Protactinium
231.04
108

109
109A
212
U
Uranium
238.03
109

110
110A
214
Np
Neptunium
237
110

111
111A
216
Pu
Plutonium
244
111

112
112A
218
Am
Americium
243
112

113
113A
220
Cm
Curium
247
113

114
114A
222
Bk
Berkelium
247
114

115
115A
224
Cf
Californium
251
115

116
116A
226
Es
Einsteinium
252
116

117
117A
228
Fm
Fermium
257
117

118
118A
230
Md
Mendelevium
258
118

119
119A
232
No
Nobelium
259
119

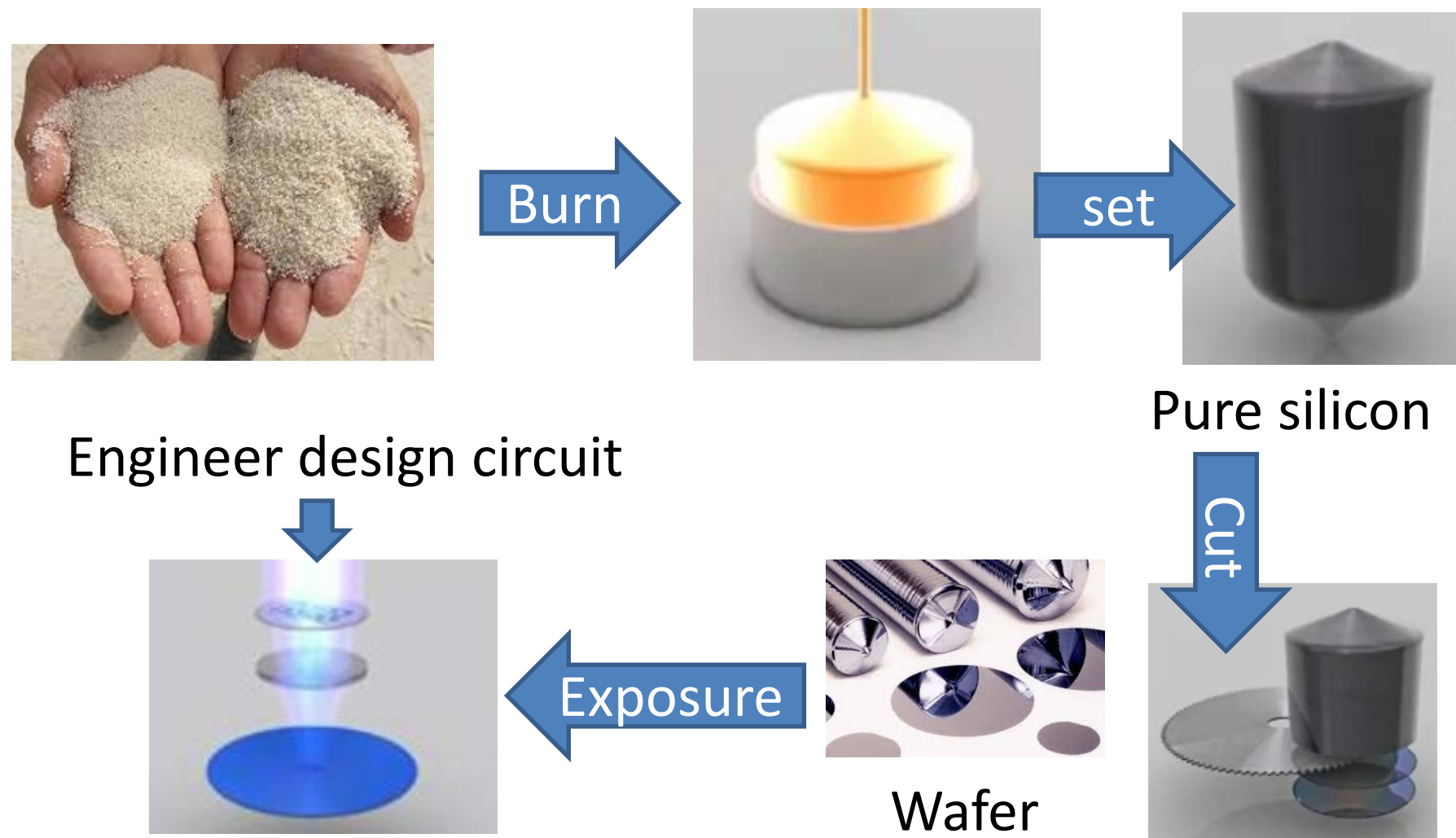
120
120A
234
Lr
Lawrencium
260
120

1.1.3 Silicon chip-making process

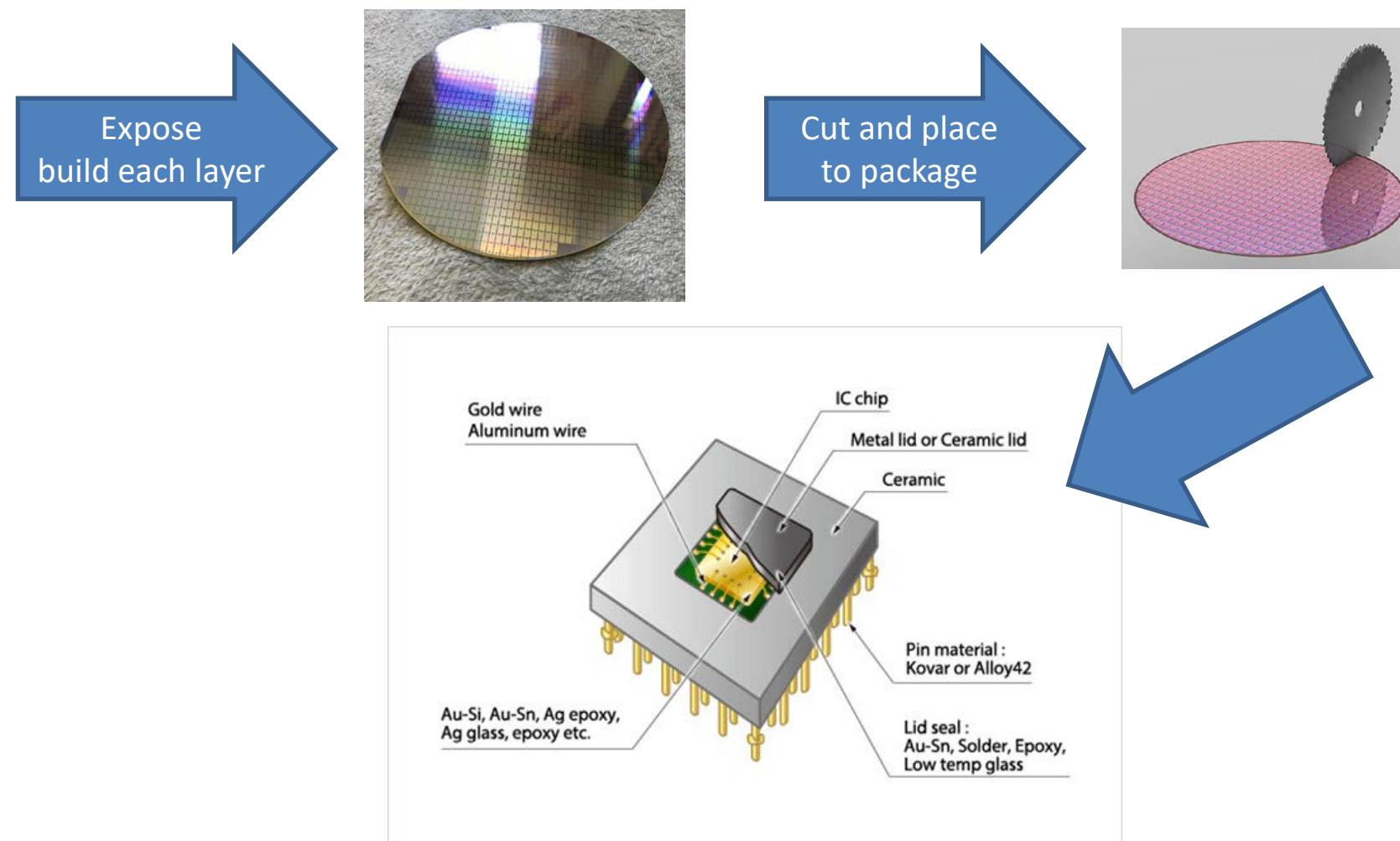
- Silicon
 - Make from the ordinary sand, quartz, rock crystal, amethyst, agate, flint, jasper, and opal.



1.1.3 Silicon chip-making process

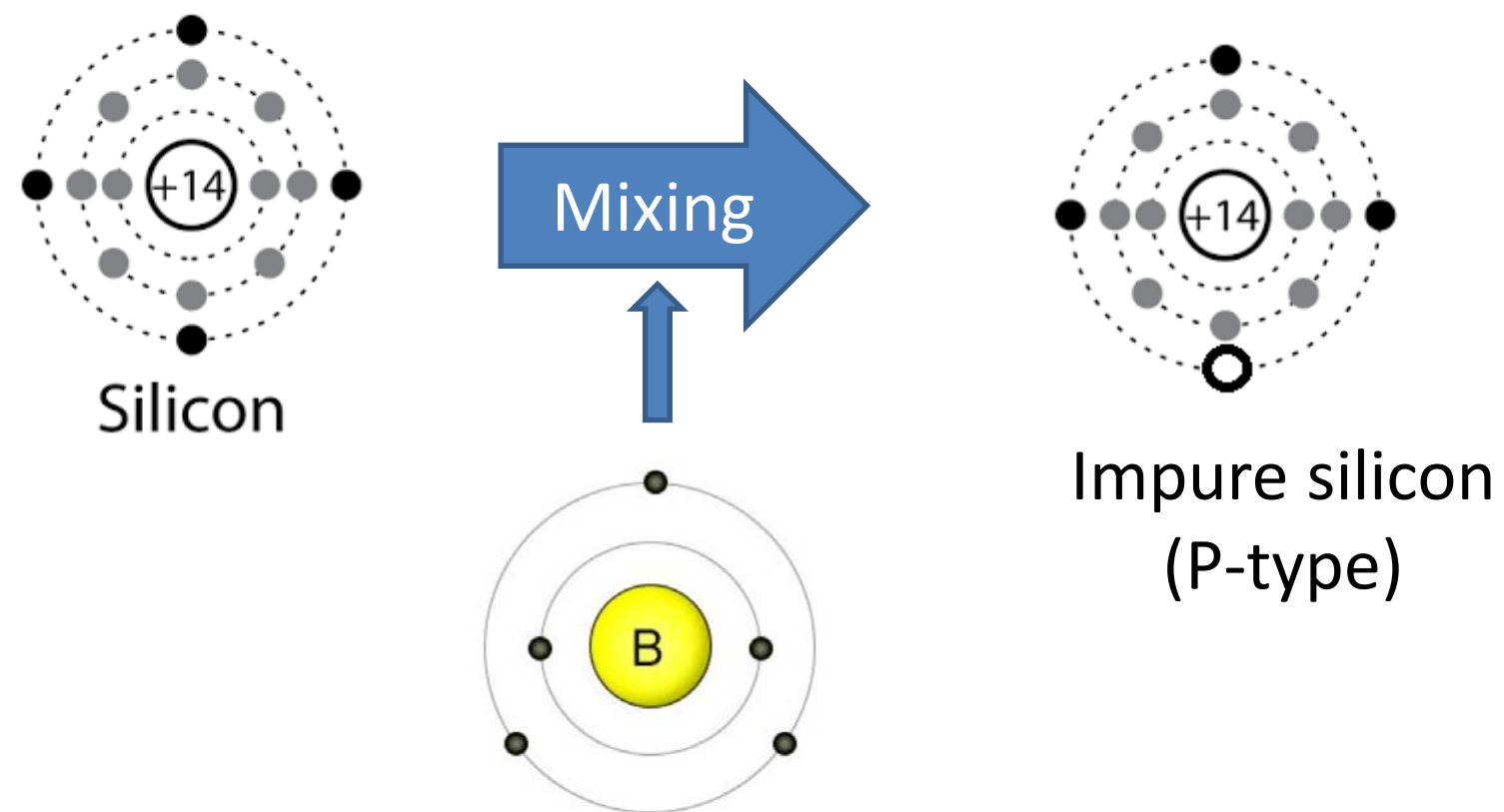


1.1.3 Silicon chip-making process



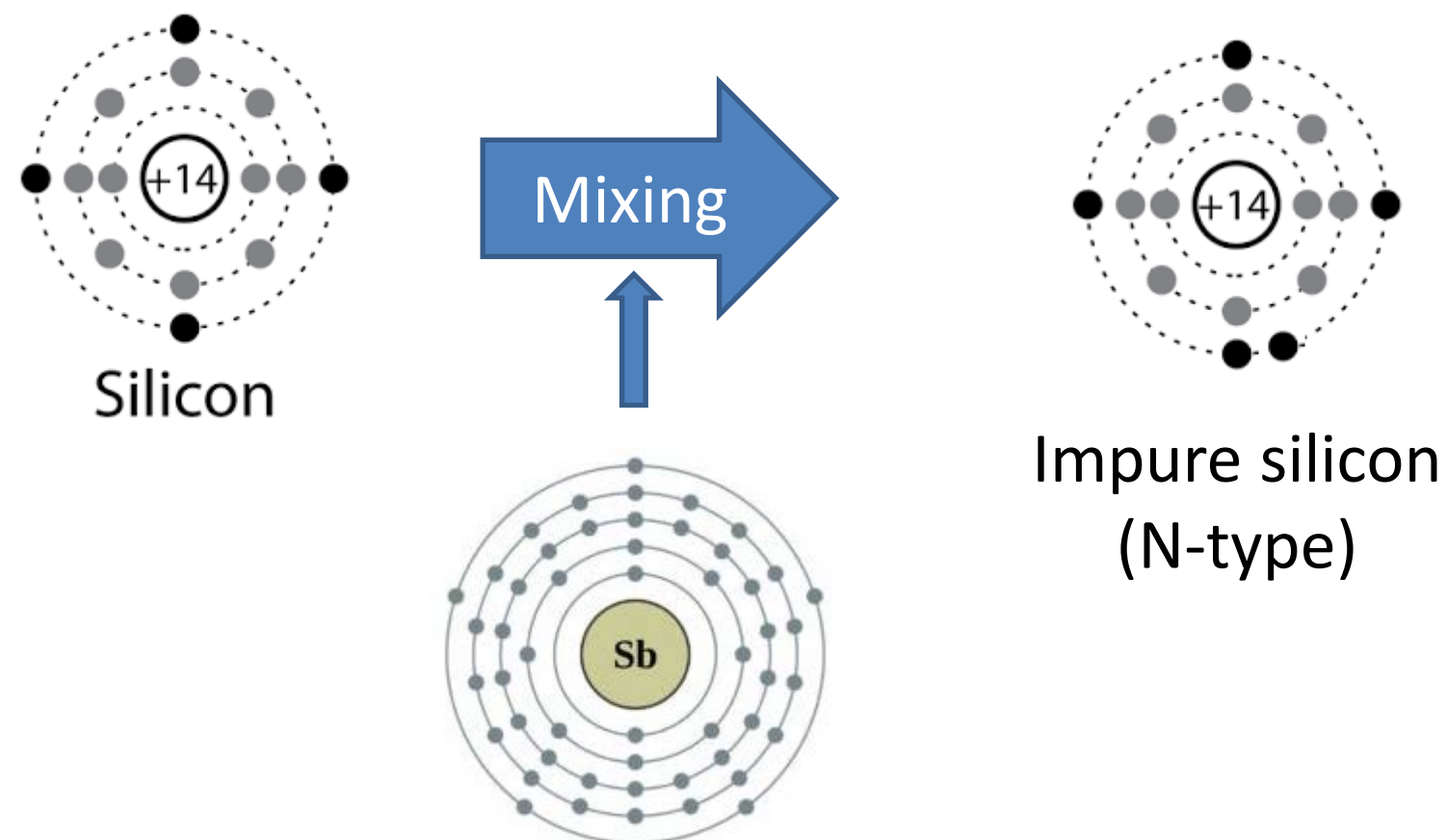
The main point of silicon chip

- Change from pure silicon to impure silicon by mixing boron to be P-type

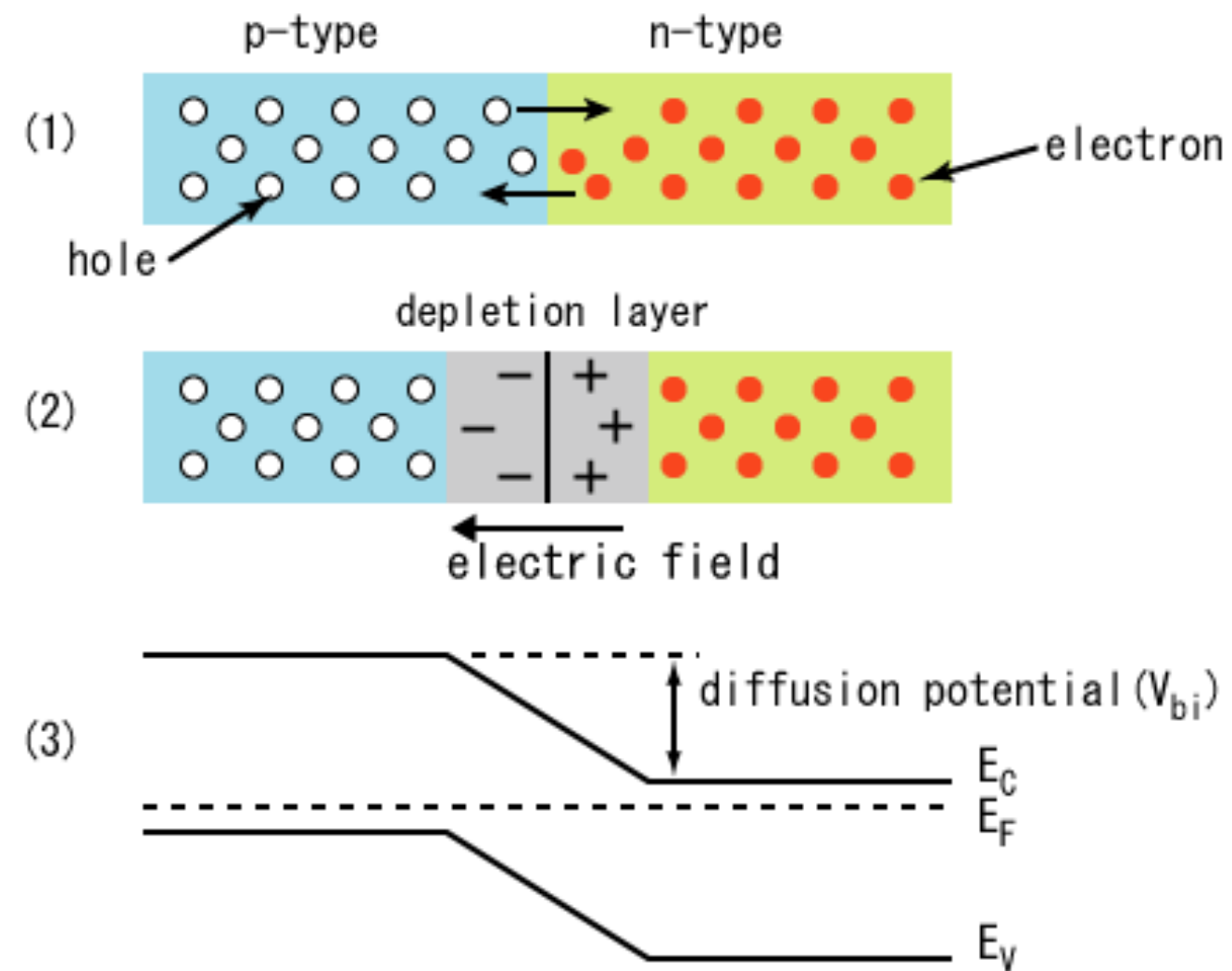


The main point of silicon chip

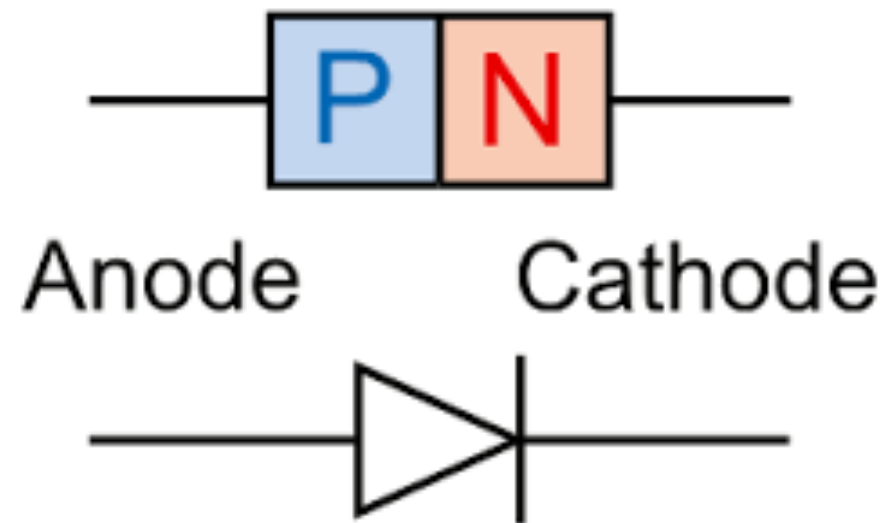
- Change from pure silicon to impure silicon by mixing antimonyแร่พวง to be N-type



N-type and P-type

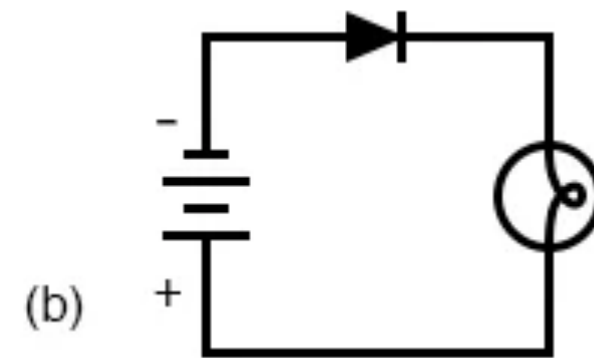
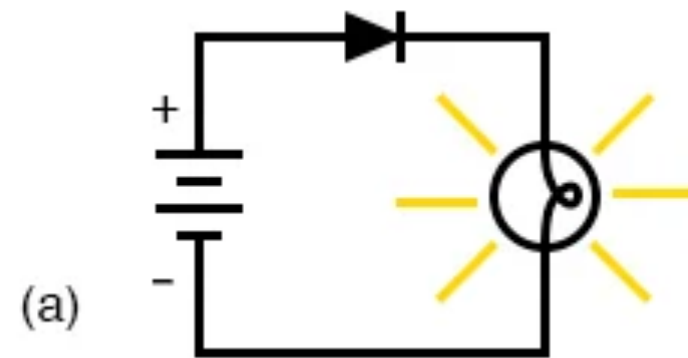


Diode

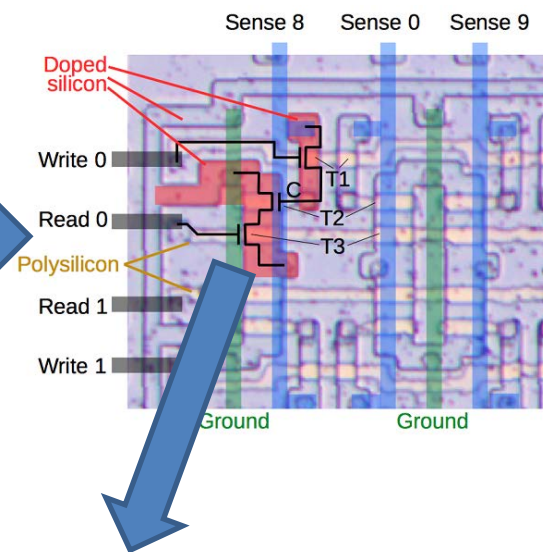
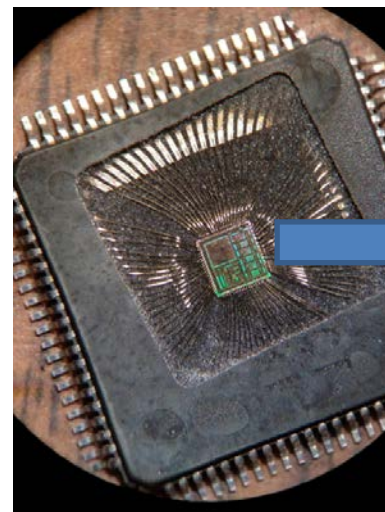


Activity 1.3

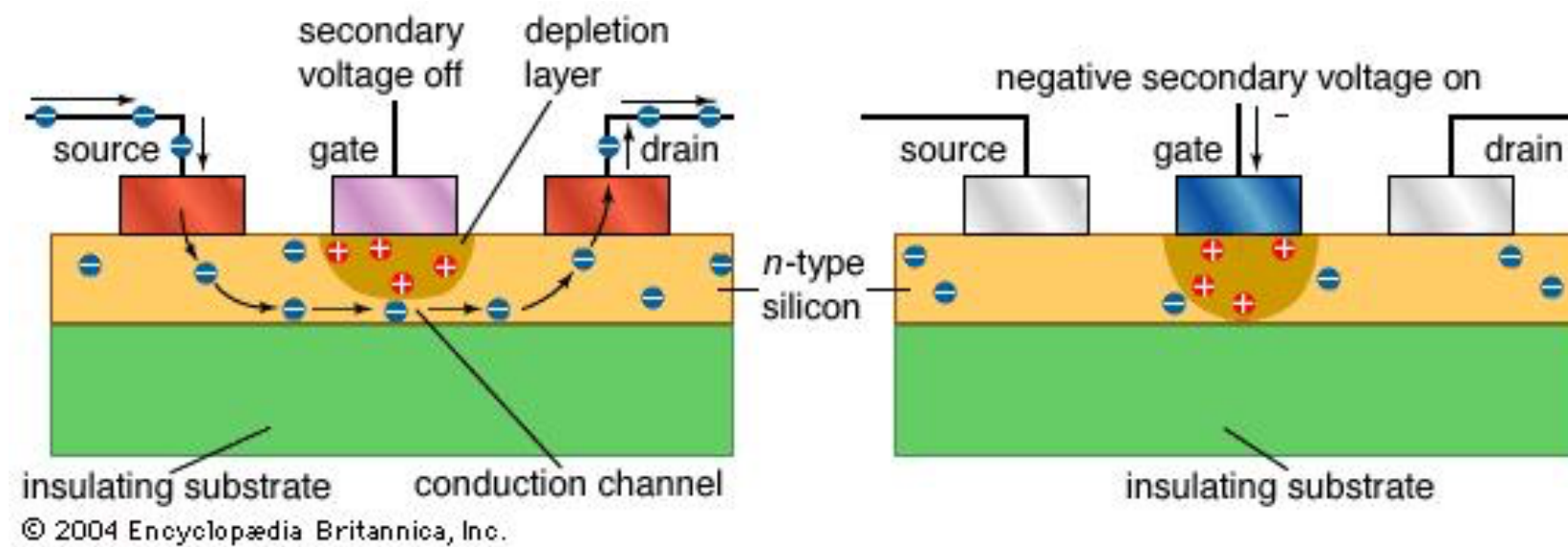
- Let's test a diode on Thinkercad.



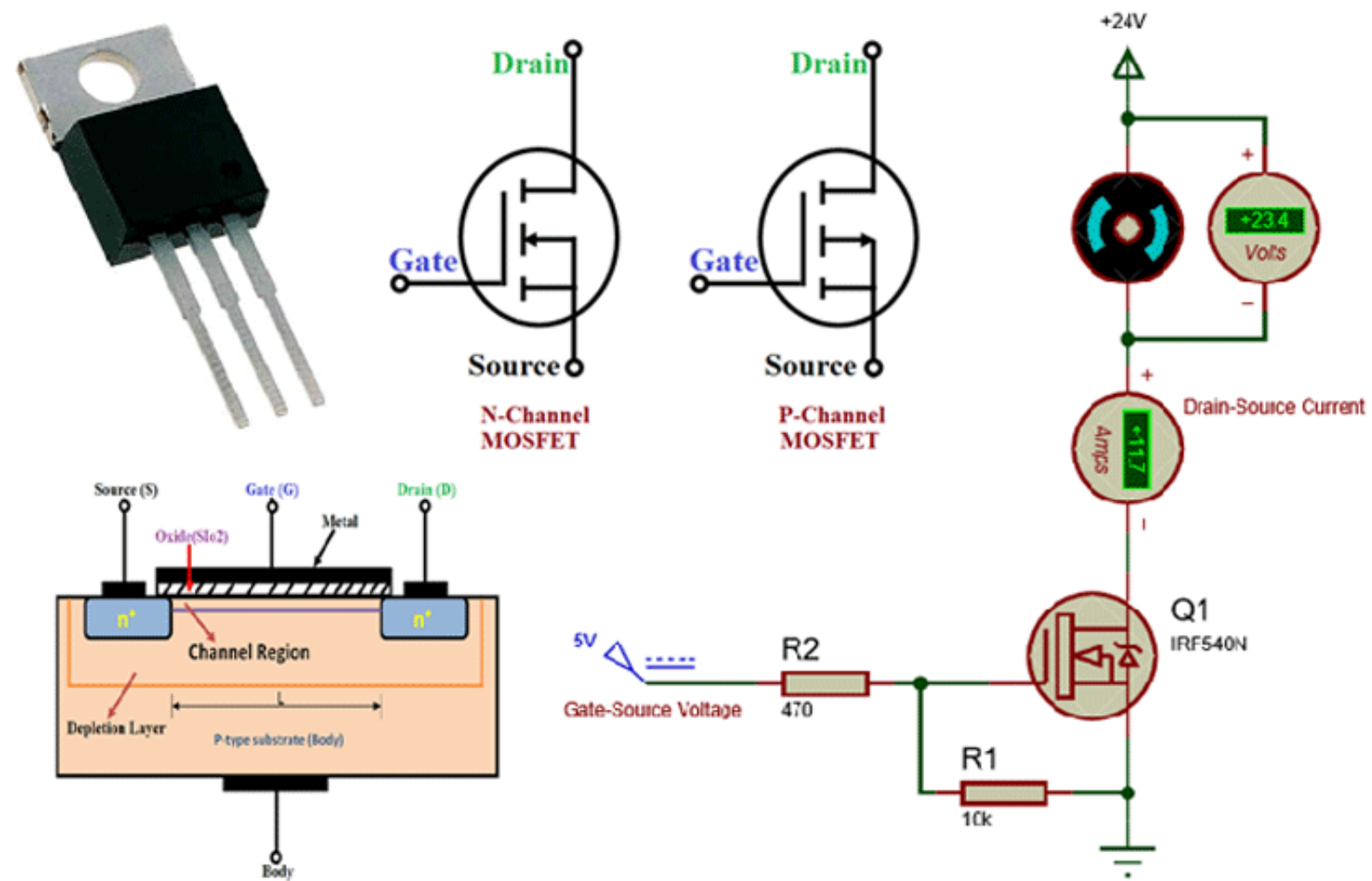
1.1.3 Silicon Transistor



- The **gate** pin controls number of the electron move from the **source** pin to the **drain**.

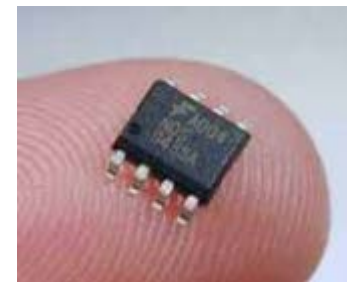
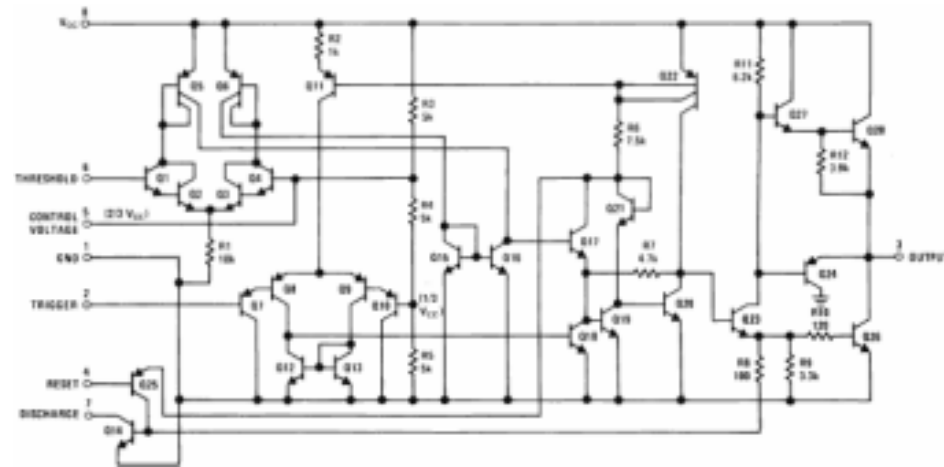


FET and Mosfet has Drain Gate Source



1.1.4 Integrated Circuit

- Discrete Circuit
- Integrated Circuit



1.1.4 Integrated Circuit: Chip Scale

Name	Number of Transistors
Small Scale IC (SSI)	10
Medium SI (MSI)	10 – 1000
Large SI (LSI)	1K – 100K
Very LSI (VLSI)	100K – 1M
Ultra LSI (ULS)	> 1M
(Nvidia GA100)	>54Bilion Transistor

Number of transistor
relate to the circuit
complexity

Year	Size of the transistor in chip
1971 - 1985	10 μm – 1 μm
1989 - 1999	800nm – 180nm
2001 - 2010	130nm – 32nm
2012 - 2017	22nm – 10nm
2018	7nm
2020	5nm
Appr. 2022	3nm

Transistor size relate to
power consumption and
switching speed

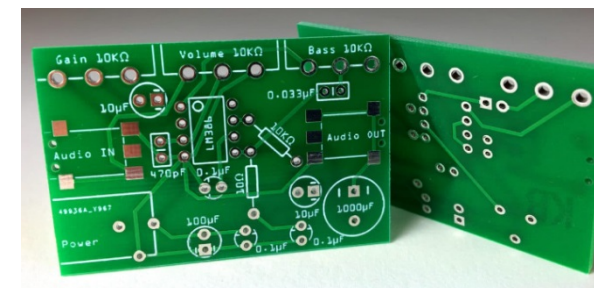
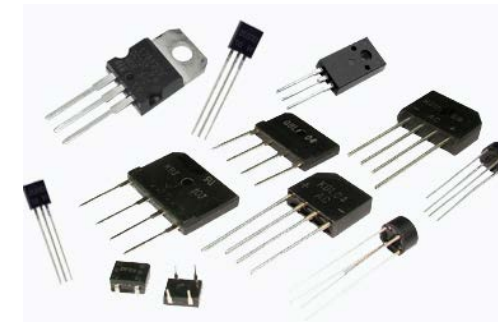
Electronic devices

- Passive device
 - Capacitor
 - Resistor
 - Inductor
 - Transformer



Electronic devices

- Active device
 - Diode
 - Transistor
 - Light Emitting Diode (LED)
 - IC, Chip
- Component
 - Print Circuit Board (PCB)



Activity 1.4

- Measure resistance and capacitance of R and C with an analog multimeter.
- Measure voltage from various battery size with an analog multimeter

1.2 The inside of Computer

- Processor
- Main memory
- System bus
- I/O Module

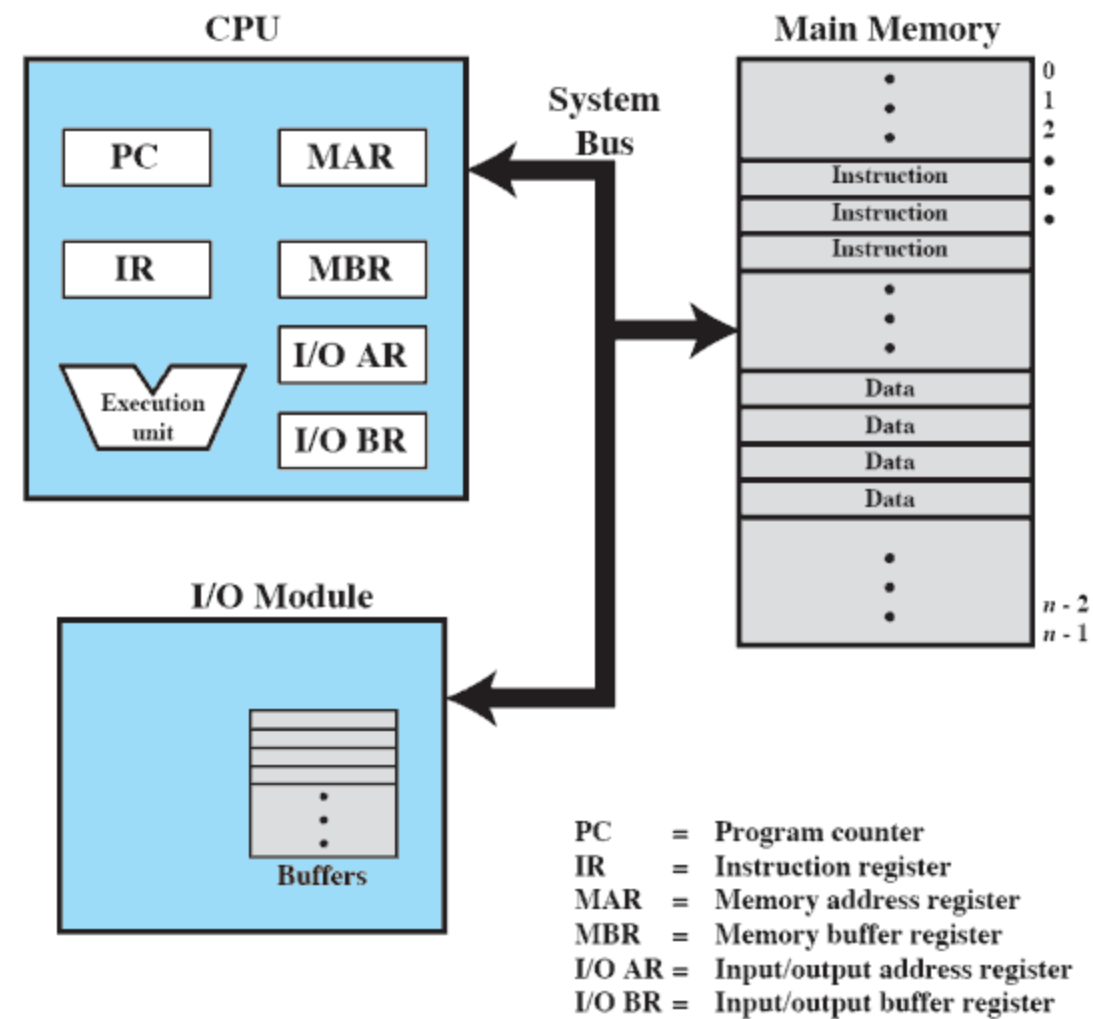
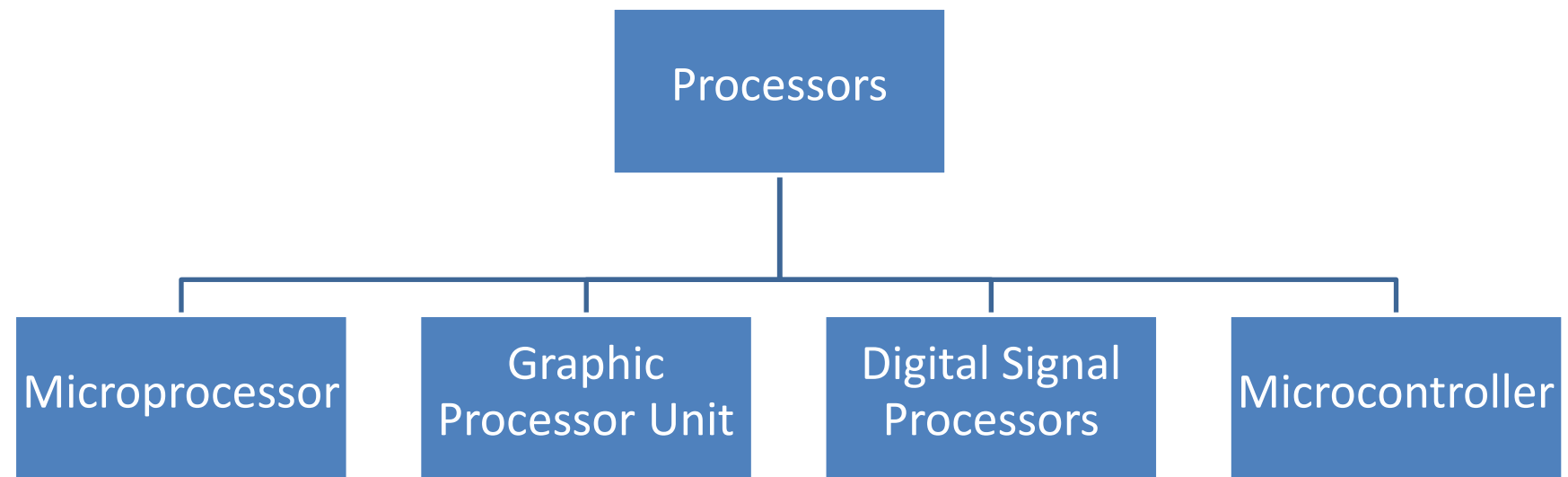
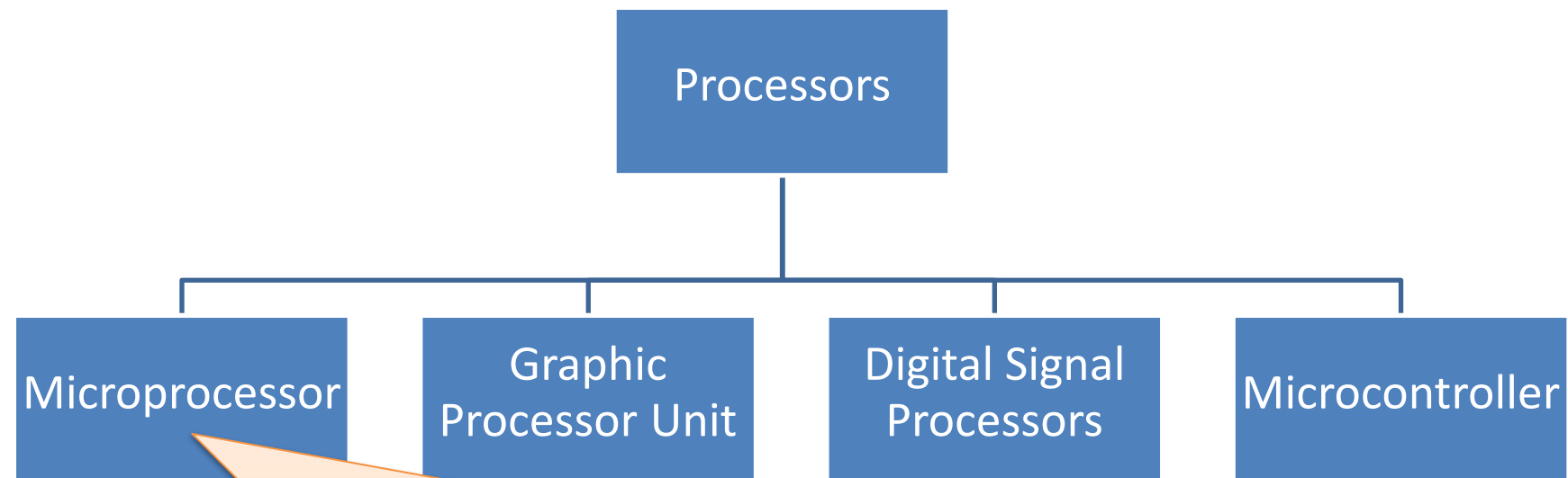


Figure 1.1 Computer Components: Top-Level View

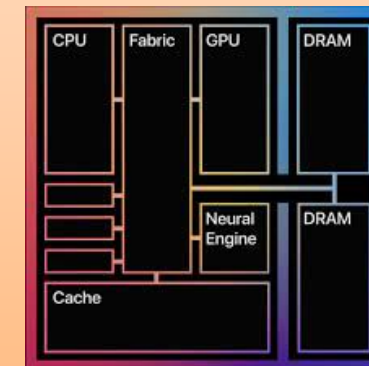
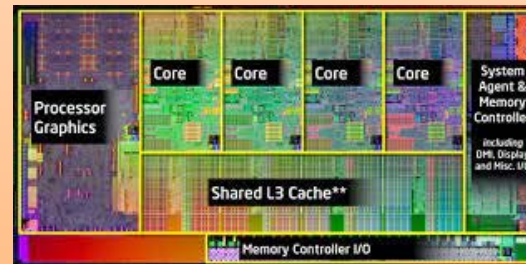




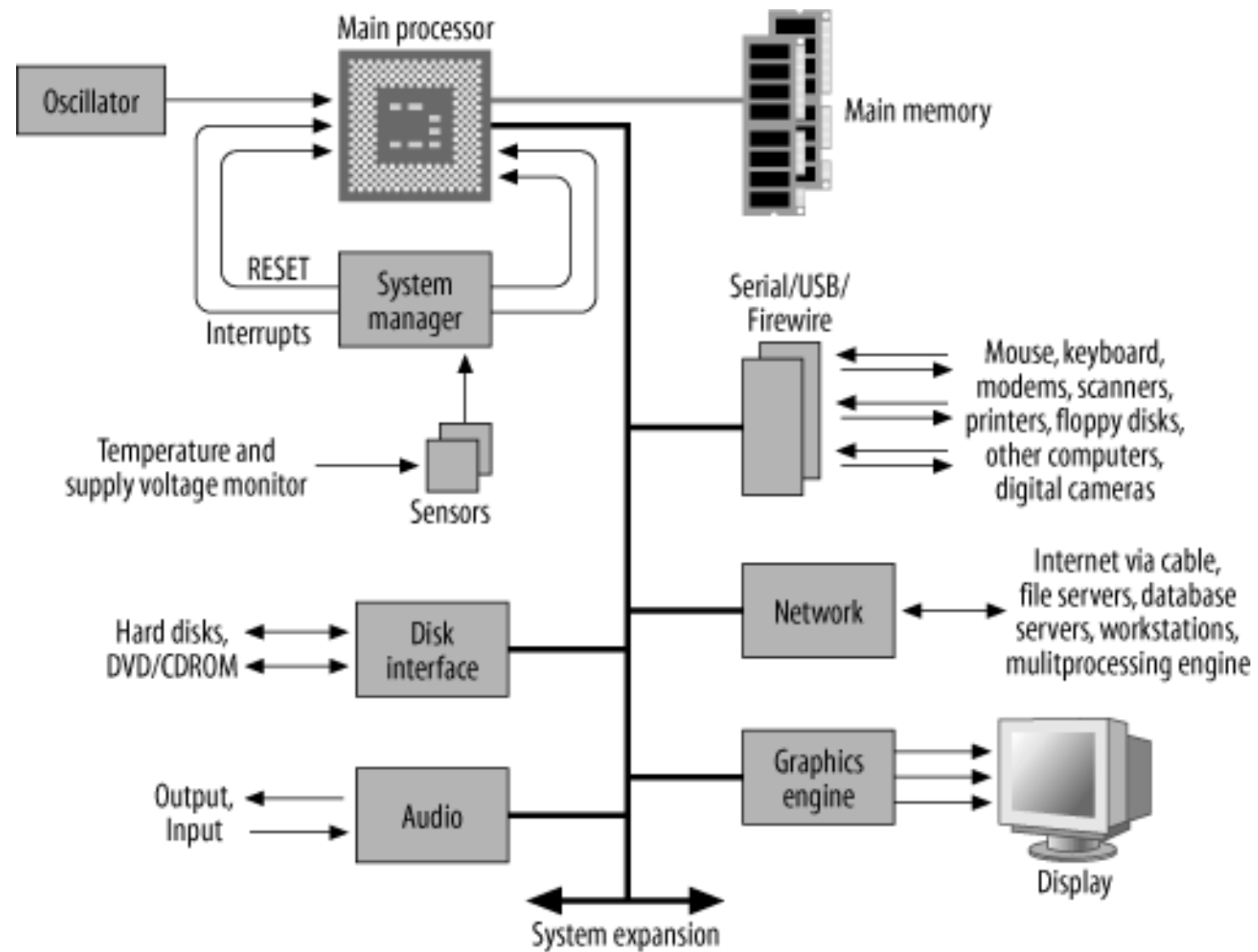
- Design instruction set for the general work
- Parallel processing
- No Input output ports (I/O)
- Some microprocessors are integrated GPU.
- Advance microprocessors are integrated GPU, I/O, and Memory by called **System-on-chip (SOC)**

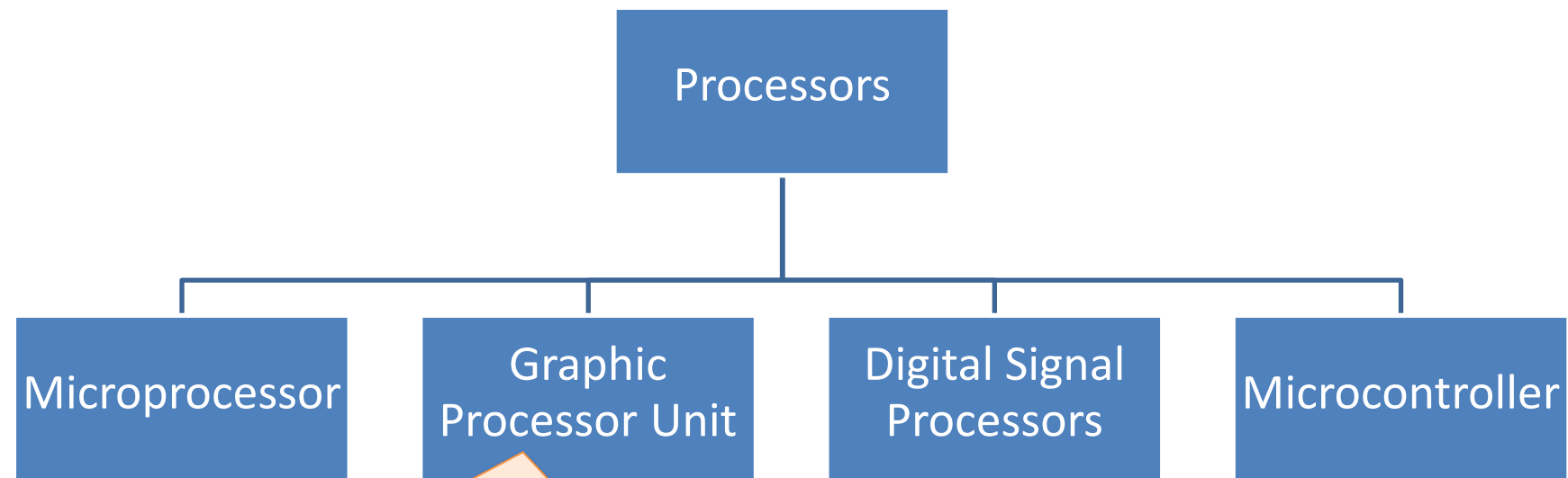
- Example

- Intel, Core-i7
- AMD, Ryzen
-

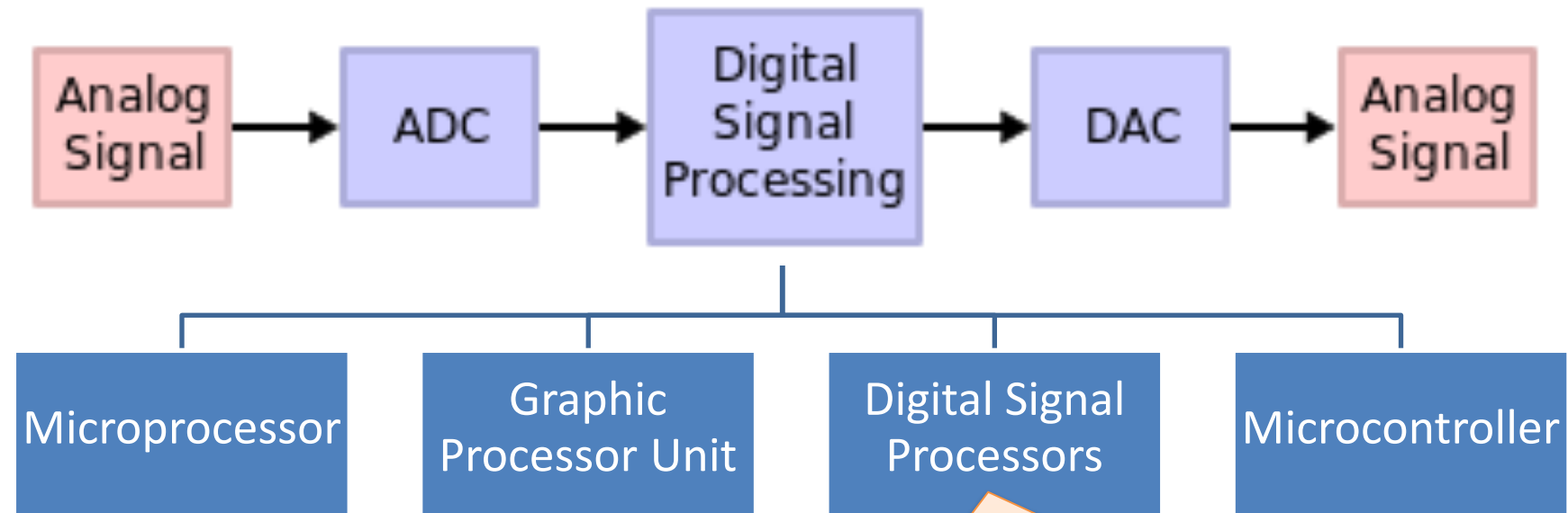


Computer systems

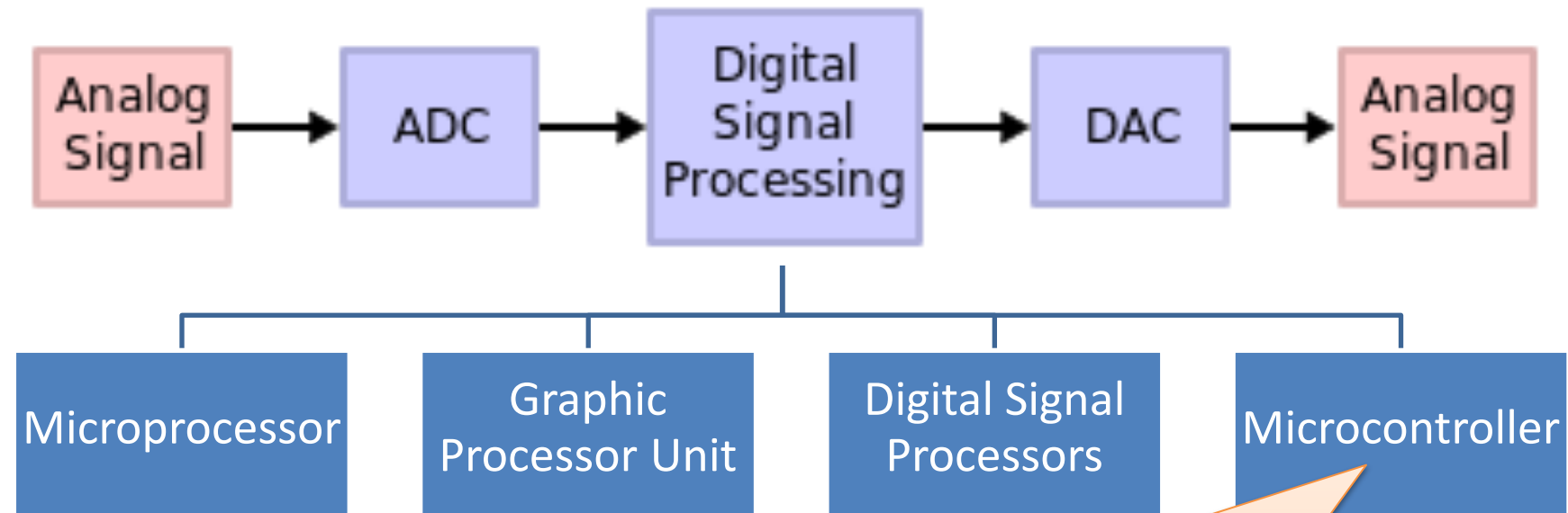




- Specific instruction set for vector and graphic computing and video rendering
- Dual ports memory interface
- Multicores
- Example family chips
 - GeForce, Radeon
 - Quadro, FirePro

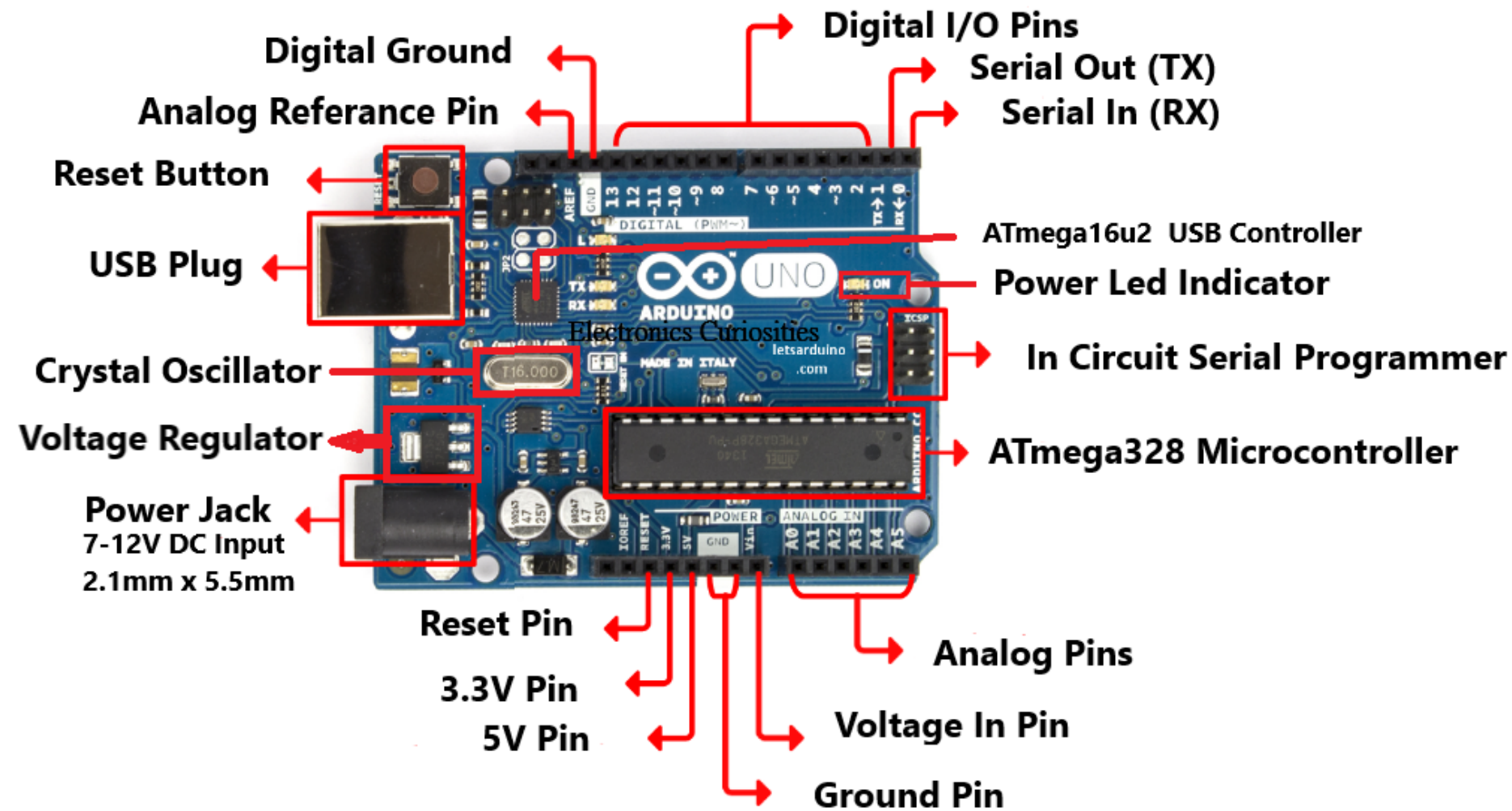


- Specific instruction set for streaming signal
 - Encoding audio/video
 - Decoding audio/video
- Example family chips
 - C6000 (Texas Instrument)
 - SHARC (Analog Devices)
 - EMU10K (Creative)



- Specific instruction set for input/output (I/O) controlling
- Design to control appliance machines such as washing machines, microwaves, toys, etc.
- Integrated I/O ports, timers, memory, Wifi, analog to digital and digital to analog.
- Example family chips
 - PIC
 - MCS-51
 - ARM

Arduino board



Activity1.5

- Identify the type of chip on the list below. You can search from Internet.
 - Z80
 - 8051
 - 68HC11
 - PIC 16Fxx
 - Atmega 328p
 - 8086
 - Geforce 1060
 - TMS32010
 - MAS3507